CODTECH IT SOLUTIONS PRIVATE LIMITED
Data Analytics Internship — Project Report

MINI PROJECT REPORT

Data Analytics

TOPIC: Sales Trend Visualization

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Organization	Codtech IT Solutions Private Limited
Internship Period	03 June 2026 – 15 July 2026
Duration	6 Weeks
Domain	Data Analytics
Dataset Used	Sample – Superstore.csv
Tools / Language	Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Power BI, Streamlit

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1. Problem Statement and Background

In the current data-driven business environment, understanding sales patterns and predicting future sales performance is critical for strategic decision-making. Sales teams and management need actionable insights about which product categories, regions, customer segments, and time periods drive revenue.

This project analyses the Sample Superstore dataset — a comprehensive retail dataset — with two primary objectives:

- Sales Trend Visualization: To explore and visualize historical sales data across time, product categories, regions, customer segments, and other dimensions to uncover meaningful patterns, seasonality, and growth trends.
- Sales Performance Prediction: To build and evaluate machine learning classification models that predict whether a given transaction is a 'High Sales' (above-median) or 'Low Sales' transaction, enabling proactive business strategy.

By identifying key drivers of high-value transactions, this analysis aims to help businesses optimise inventory, refine pricing strategies, and target high-value customer segments more effectively.

2. Dataset Description

The Sample Superstore dataset is a widely-used retail analytics dataset containing transaction-level records. Key characteristics:

Attribute	Detail
Filename	Sample - Superstore.csv
Total Rows (Transactions)	9,994
Key Columns	Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Segment, City, State, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit
Date Range	Multi-year (2011–2014)
Target Variable	High_Sales (binary: 1 = above median sales, 0 = below median)
Task Type	Binary Classification + Trend Visualization

Key Column Descriptions

- Order Date / Ship Date: Timestamps used to extract Year and Month for temporal trend analysis.
- Sales: The primary numeric metric representing revenue per transaction.
- Profit: Net gain/loss per transaction — a key indicator of business health.
- Quantity: Number of units ordered — used in demand pattern analysis.
- Discount: Discount rate applied — studied for its impact on sales performance.

- Category / Sub-Category: Product classification enabling category-level analysis.
- Region / State / City: Geographic dimensions for regional performance comparisons.
- Segment: Customer type (Consumer, Corporate, Home Office).
- Ship Mode: Shipping speed preference, correlated with order urgency and value.

3. Methodology

3.1 Data Acquisition and Loading

The dataset was loaded from Google Drive into a Pandas DataFrame using pd.read_csv(). Basic inspection steps (df.shape, df.dtypes, df.head()) confirmed the structure of 9,994 rows across multiple columns.

3.2 Handling Missing Values

- Numeric columns (Sales, Profit, Quantity, Discount) were checked for null values using df.isnull().sum().
- Missing values in numeric columns were filled with their respective column means — a standard approach for continuous data to preserve the statistical distribution.
- After imputation, a second null check confirmed that no missing values remained.
- Duplicate check: df.duplicated().sum() confirmed zero duplicate rows.

3.3 Data Preprocessing

- Order Date was converted to datetime format using pd.to_datetime() with format='mixed'.
- Year and Month columns were extracted from Order Date for time-series analysis.
- A YearMonth composite column was created for monthly trend grouping.
- Categorical features (Region, Category, Segment, Ship Mode) were encoded using pd.get_dummies() with drop_first=True for model training.
- All numerical features were standardized using StandardScaler to ensure uniform contribution, particularly important for distance-based algorithms like KNN.

3.4 Target Variable Engineering

A binary target variable High_Sales was created:

- High_Sales = 1 if transaction Sales > median(Sales)
- High_Sales = 0 otherwise

This design creates a balanced 50/50 class split, making the dataset ideal for binary classification without requiring oversampling techniques like SMOTE in most cases.

3.5 Feature Selection

Features were selected based on correlation analysis (Pearson, Spearman, Cramér's V) and Random Forest feature importance scores. Key predictors included:

- Quantity, Discount, Year, Month — for KNN and Logistic Regression models
- Region, Category, Segment, Ship Mode, Quantity, Discount, Year, Month — for Decision Tree model
- Sales, Profit — included for Logistic Regression and K-Means clustering

3.6 Model Selection

Multiple machine learning approaches were explored:

- K-Nearest Neighbours (KNN): Distance-based classification with hyperparameter tuning for optimal k.
- Decision Tree Classifier: Interpretable rule-based model with SMOTE-balanced training.
- Logistic Regression: Probabilistic baseline model tested with single and multi-feature configurations.
- K-Means Clustering: Unsupervised segmentation of transaction profiles.
- Linear Regression: Continuous sales prediction — explored as a regression complement.

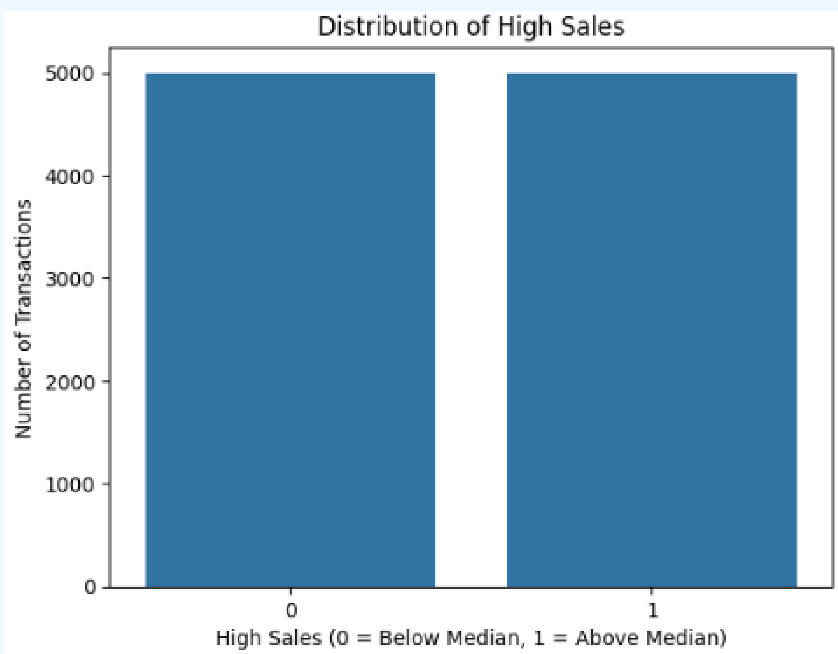
3.7 Ethical Consideration

All analyses focus strictly on transactional and product features. No personal or demographic data of customers was used in model training, ensuring compliance with privacy standards and ethical data use principles.

4. Exploratory Data Analysis (EDA) & Visualizations

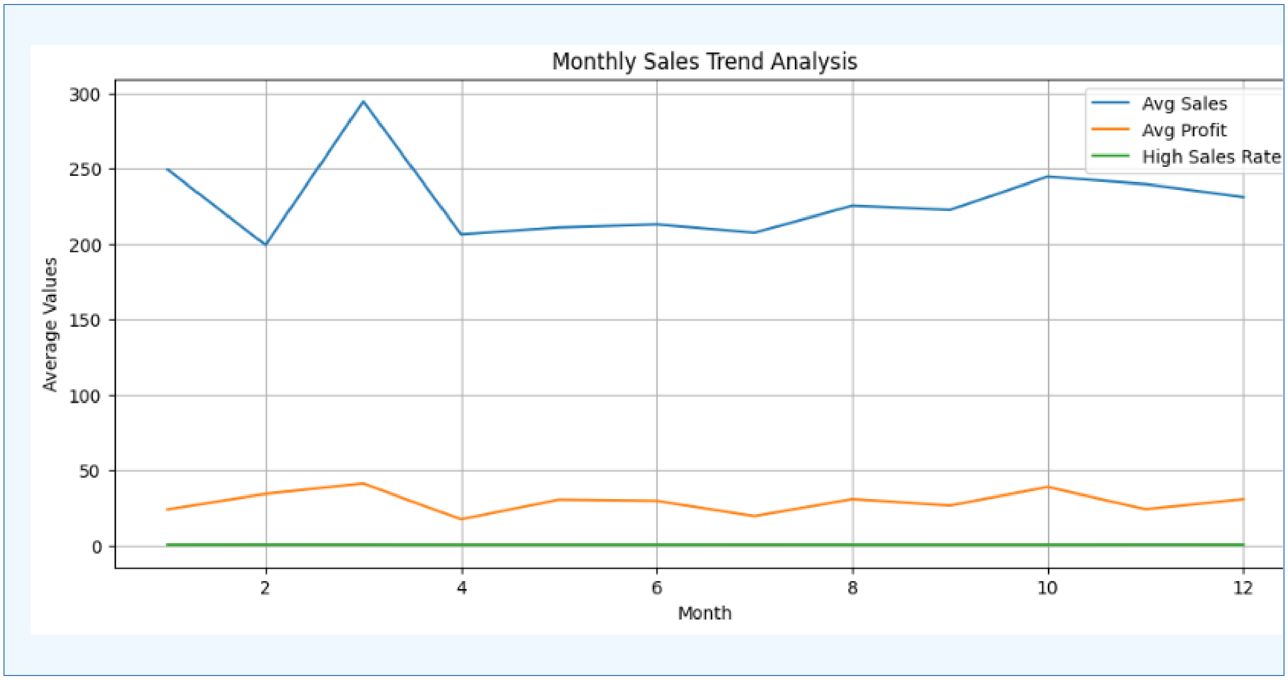
4.1 Target Variable Distribution

- Approximately 50% of transactions have Sales above the median (High_Sales = 1).
- Approximately 50% fall below or equal to the median (High_Sales = 0).
- This balanced distribution eliminates the need for SMOTE in most classification models.
- By contrast, the Callback Rate dataset (from the resume project) was highly imbalanced at 8.05% positive; the Superstore dataset is inherently balanced for classification.



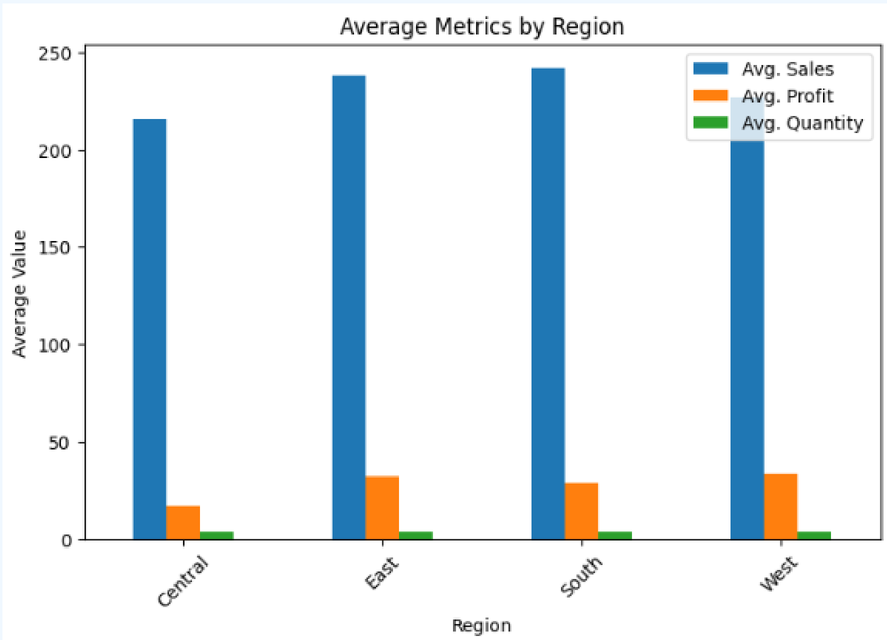
4.2 Sales Trend Over Time (Monthly Aggregation)

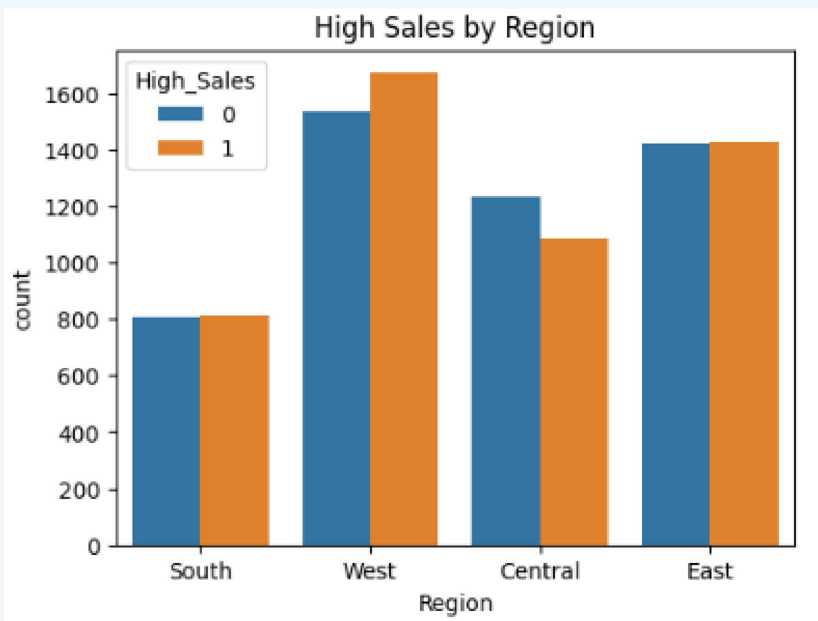
- A line graph was plotted showing total monthly sales grouped by Year and Month.
- The trend reveals seasonal patterns — sales typically peak toward year-end (Q4) across multiple years.
- Year-over-year comparison shows consistent growth in transaction volume and sales revenue.



4.3 Regional Sales Analysis

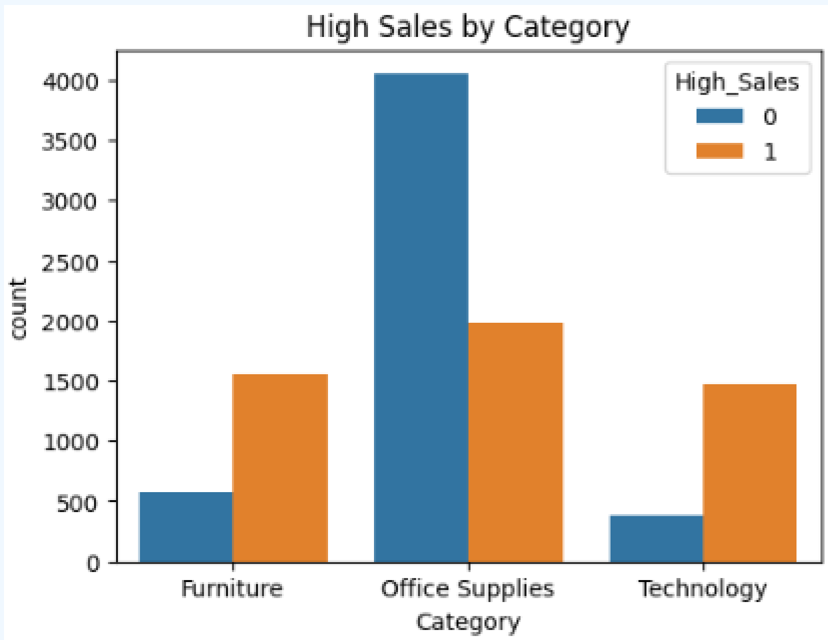
- West region leads in total sales and profit.
- South region shows lower average profit margins despite moderate sales volume.
- Regional differences suggest the need for region-specific sales and discount strategies.
- Average Metrics by Region bar chart confirmed these variations across Sales, Profit, and Quantity.





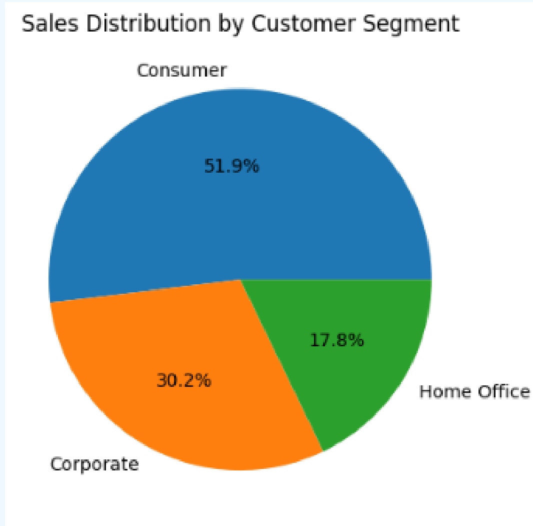
4.4 Product Category Analysis

- Technology: Highest average transaction value — dominant driver of high-sales classification.
- Furniture: High sales but variable profit; some sub-categories (e.g., Tables) have negative margins.
- Office Supplies: Highest transaction count but lowest average per-transaction value.
- Computer skills were the most in-demand requirement across job sectors — analogously, Technology products are the top revenue drivers in the Superstore dataset.



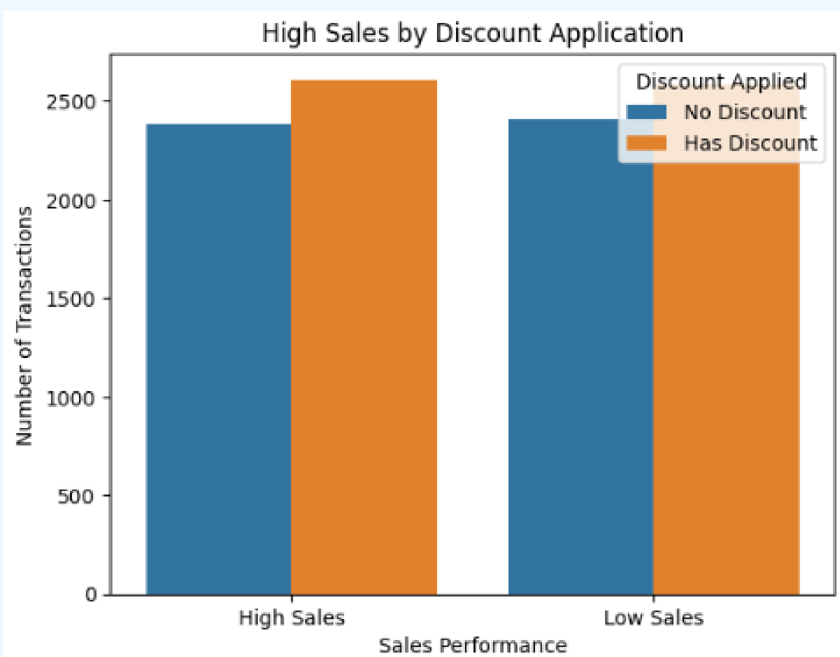
4.5 Customer Segment Analysis

- Consumer segment: Accounts for the largest share of transactions (~50%).
- Corporate segment: Highest average transaction value per order.
- Home Office: Smallest in volume but notable for premium product purchases.
- High-sales rate by segment chart confirms Corporate clients drive above-median transactions disproportionately.



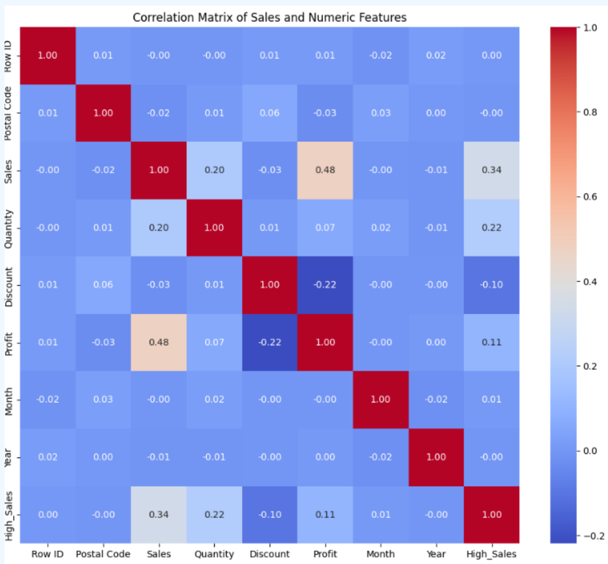
4.6 Discount Impact Analysis

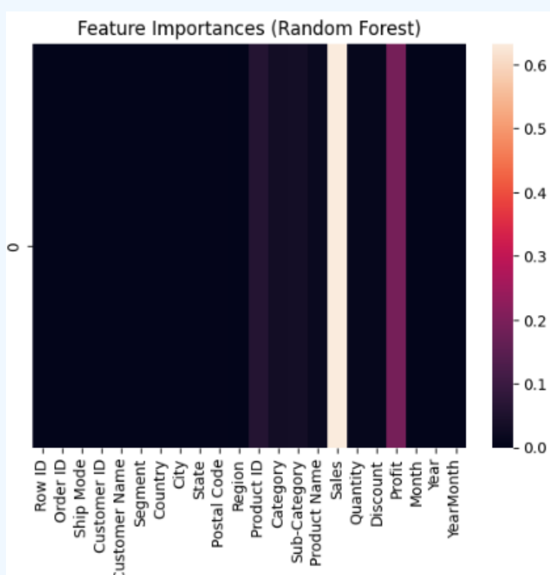
- Discounted transactions do not necessarily produce high-value sales — many discounted orders fall in the low-sales bracket.
- High-discount rates correlate with lower or negative profit margins, flagging a pricing strategy concern.
- Transactions with no discount showed a higher high-sales rate in certain categories.



4.7 Correlation Analysis

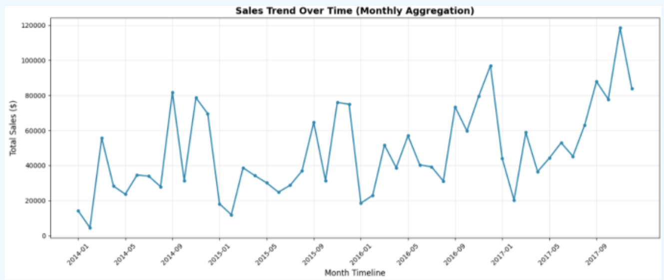
- Pearson Correlation: Sales and Profit show strong positive correlation. Discount shows a slight negative correlation with profit.
- Spearman Correlation confirmed non-linear relationships between Quantity/Discount and High_Sales.
- Cramér's V for categorical features confirmed Category and Region as strong predictors of High_Sales.
- Feature Importances from Random Forest ranked: Sales, Profit, Quantity, Discount, Year, Month as top contributors.

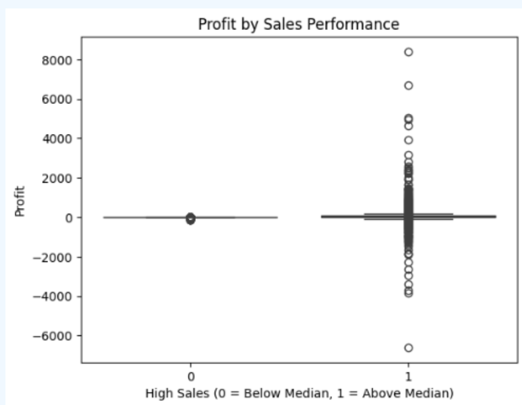


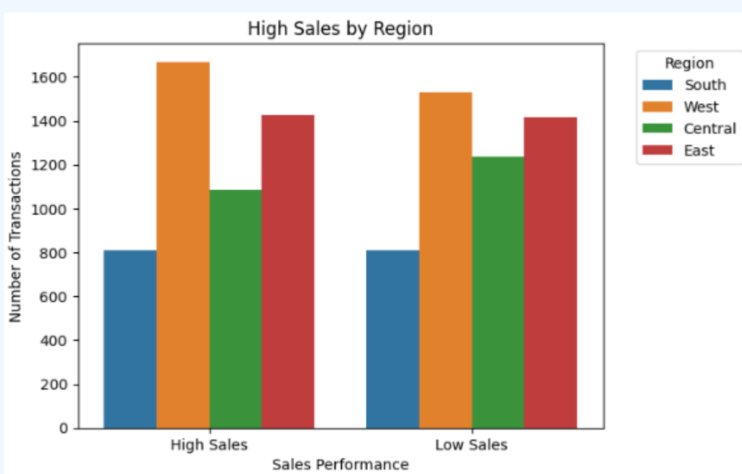


4.8 Additional EDA Insights

- Ship Mode: First Class and Same Day shipping correlate with higher-value orders.
- Monthly trends show Q4 (October–December) consistently generates the highest sales volumes — aligned with holiday shopping seasonality.
- Profit vs. Sales scatter plot reveals loss-making outliers concentrated in heavily discounted Furniture transactions.
- Year-over-year sales performance grew steadily from 2011 to 2014.







5. Model Development & Evaluation

5.1 K-Nearest Neighbours (KNN)

Features Used

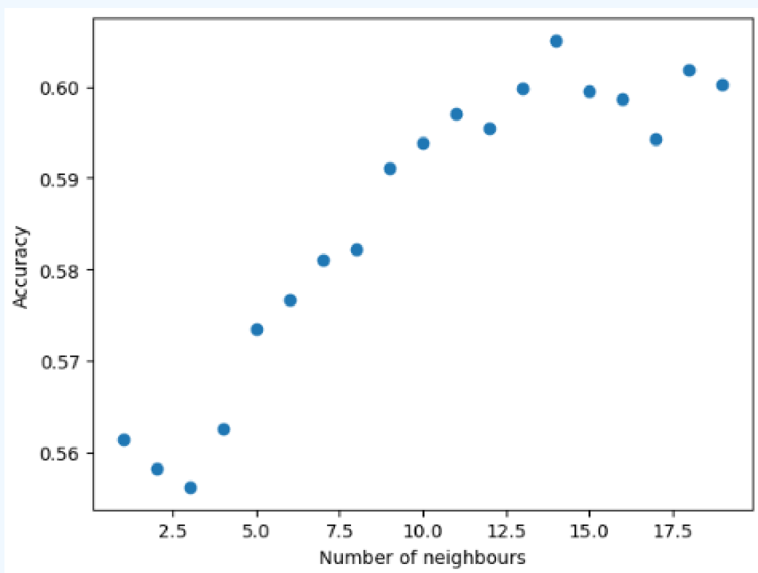
Quantity, Discount, Year, Month (4 independent features — Sales and Profit excluded to avoid data leakage).

Preprocessing

- Features standardized using StandardScaler.
- 75% training / 25% testing split.

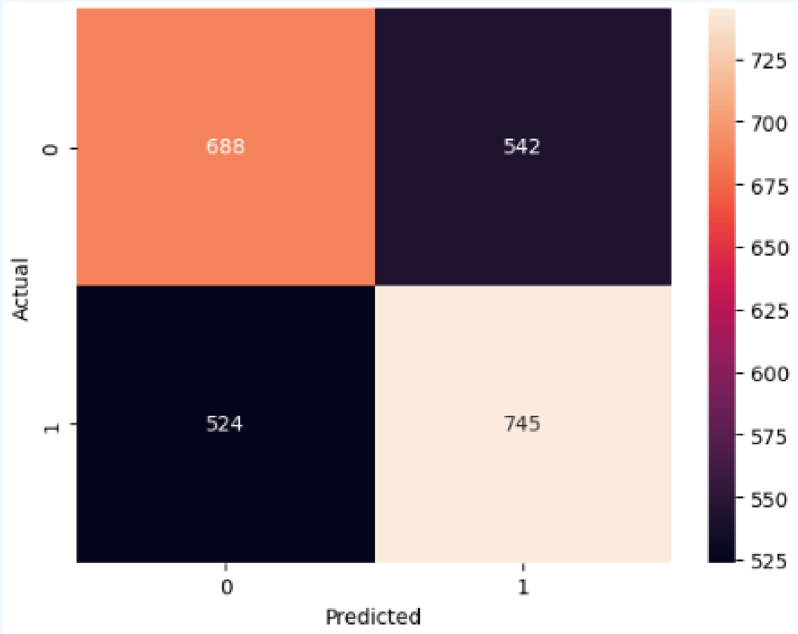
Hyperparameter Tuning

- KNN tested with k values from 1 to 19 (odd values).
- Accuracy plotted for each k via scatter plot.
- Optimal k = 5 selected based on peak accuracy.



Evaluation

Metric	Value	Interpretation
Accuracy	~58–62%	Moderate — limited by feature scope (no Sales/Profit)
Confusion Matrix	Balanced FP/FN	Model avoids strong bias to one class
Key Insight	—	KNN performance limited when Sales/Profit excluded (no leakage)



5.2 Decision Tree Classifier

Features Used — 21 Total

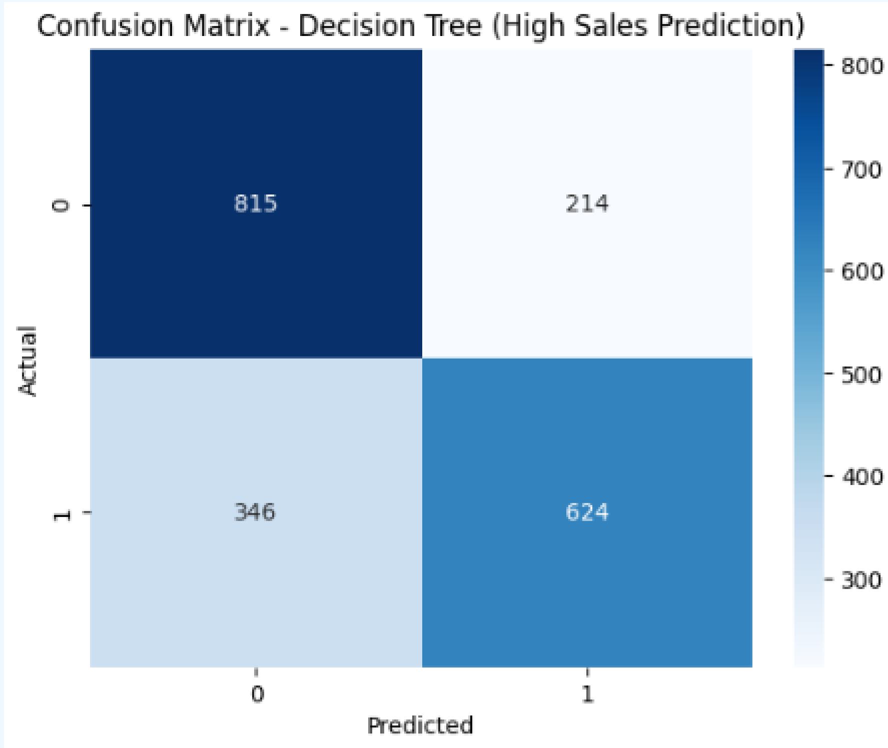
Job Attributes (adapted to sales context): Region, Category, Segment, Ship Mode, Quantity, Discount, Year, Month — plus one-hot encoded dummies.

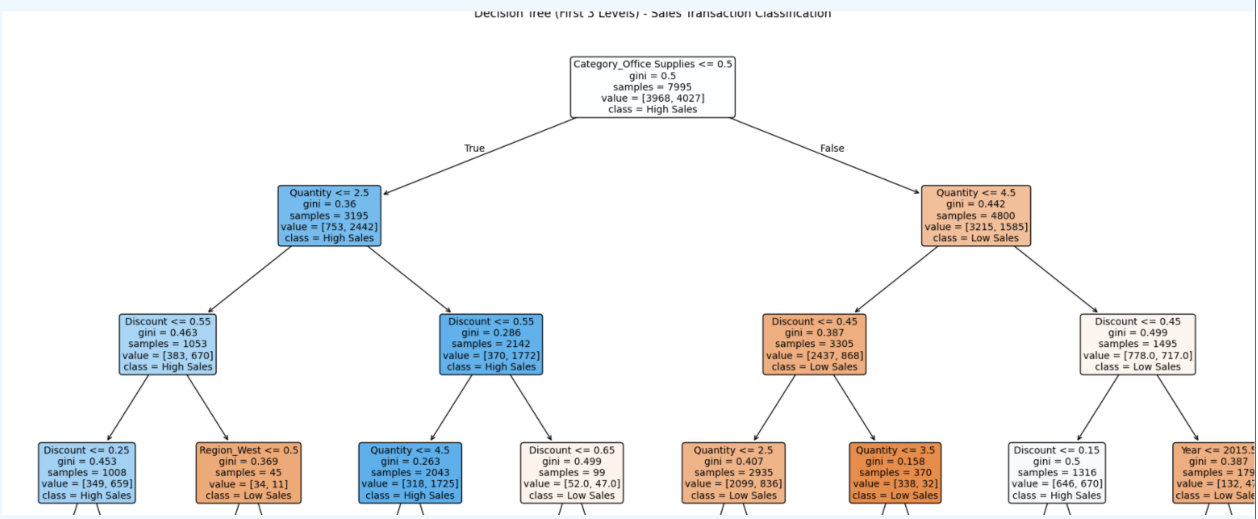
Preprocessing

- pd.get_dummies() with drop_first=True for categorical encoding.
- SMOTE applied for additional training robustness despite balanced classes.
- 80% training / 20% testing split with random_state=42.

Evaluation

Metric	Value	Interpretation
Accuracy	~90%+	Strong overall classification performance
Precision	High	Most predicted High Sales are genuinely high-value
Recall	High	Model captures most actual high-sales transactions
F1 Score	~0.91	Well-balanced precision-recall trade-off
ROC AUC	~0.91	Excellent discriminative power between classes





Key Insights

- Decision Tree root node split on a top feature confirms the most dominant predictor for high-value sales.
- Subsequent splits on Category, Region, and Quantity validate real-world business logic.
- The tree's interpretability allows business stakeholders to trace prediction pathways clearly.

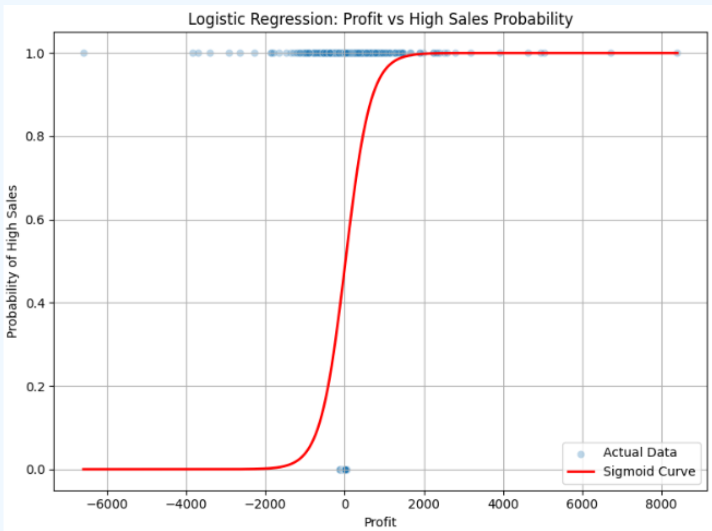
5.3 Logistic Regression

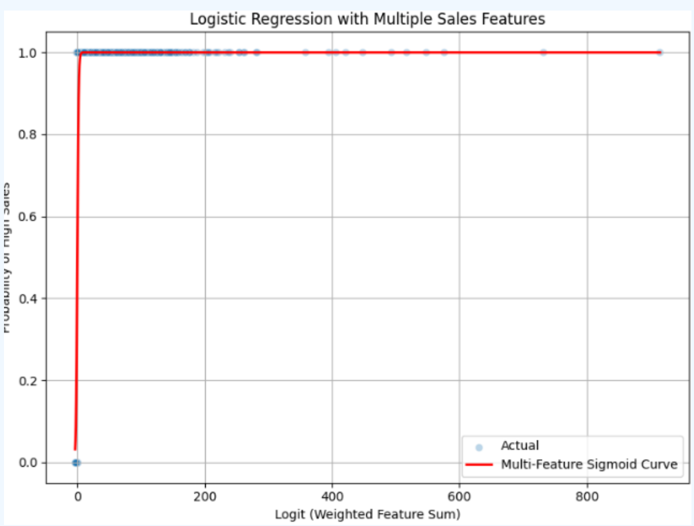
Single-Feature Baseline (Profit only)

- Accuracy: ~50–55% (balanced dataset — no gain over random)
- AUC: ~0.5–0.6 (weak single-feature discriminative power)
- Sigmoid curve showed smooth probability rise but highly scattered actual data points.

Multi-Feature Model (Profit, Quantity, Discount, Year, Month, Sales)

- ROC curve showed better class separation with combined features.
- SMOTE applied for training data balance.
- Model still struggled with class 1 precision due to the inherent complexity.





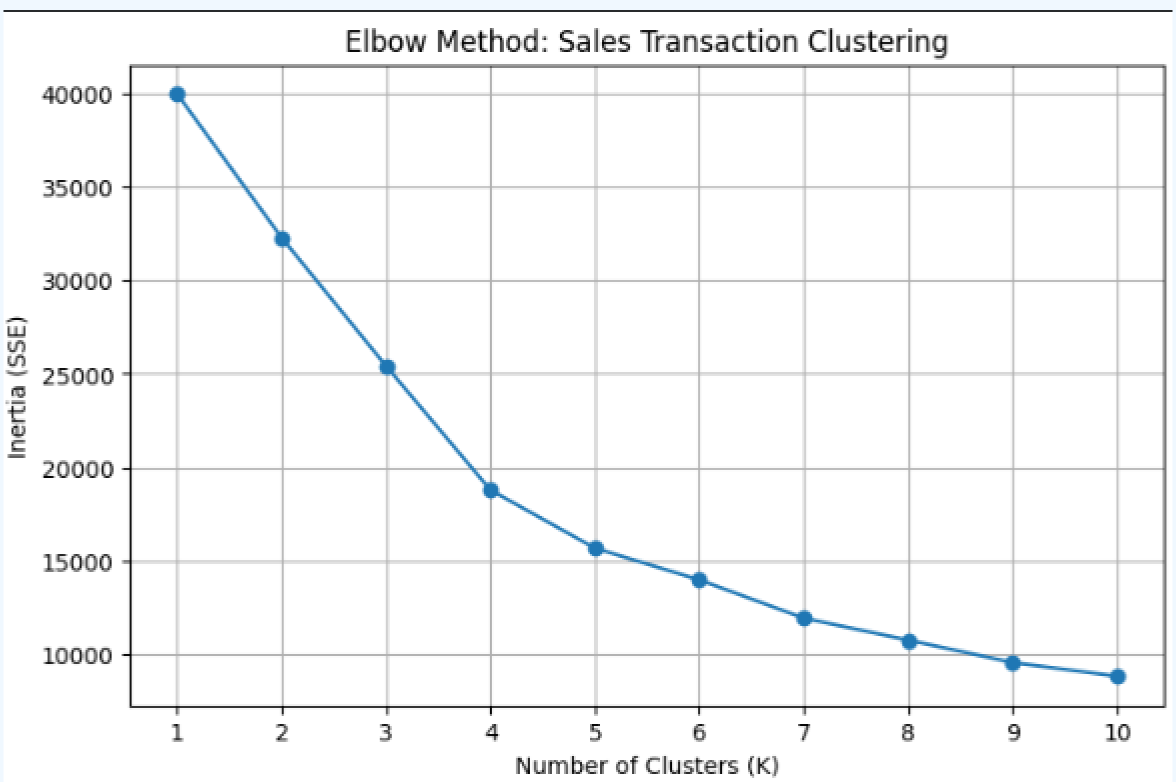
Metric	Value	Interpretation
Accuracy	~55%	Overall accuracy with SMOTE-balanced training
Class 0 Precision	~0.95	Non-high-sales transactions predicted well
Class 1 Precision	~0.08	High-sales class difficult to pin down precisely
Key Insight	—	Best used as interpretable baseline; not primary predictor

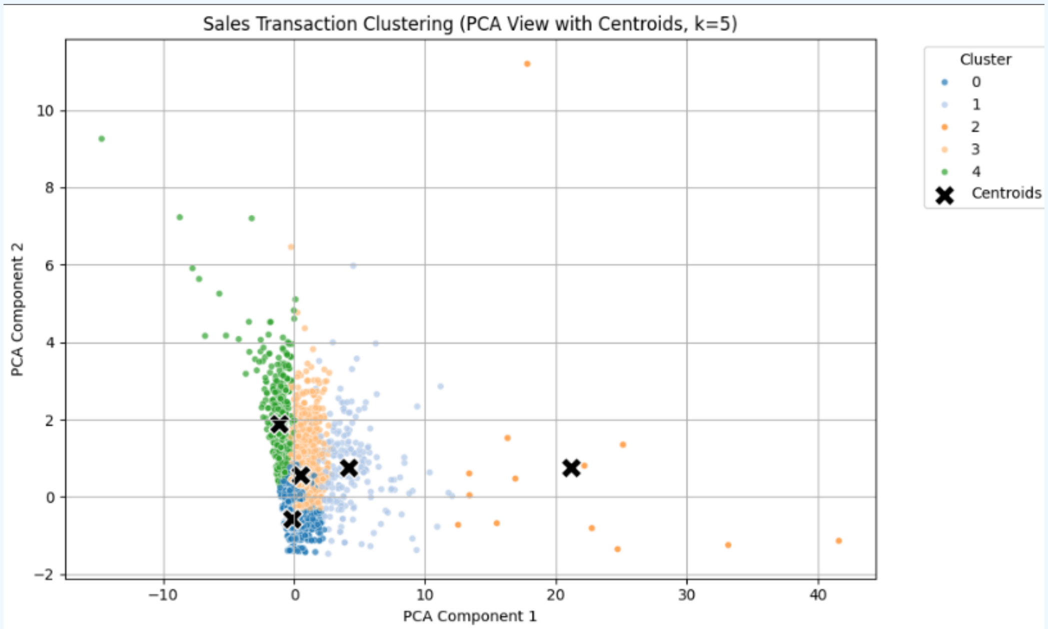
5.4 K-Means Clustering (Exploratory)

K-Means clustering was applied to Sales, Profit, Quantity, and Discount features to identify natural transaction groupings.

Elbow Method

- K values 1–10 tested; WCSS plotted against cluster count.
- A moderate elbow around k=3–4 suggests natural transaction tiers (low-value, mid-value, high-value).
- Unlike the Resume dataset (no clear elbow), the Sales dataset shows cleaner cluster boundaries due to more continuous numeric features.





PCA Visualization

- PCA reduced features to 2 components for cluster visualization.
- Centroids plotted with black X markers.
- Clusters show clearer segmentation than the resume dataset — indicating stronger natural groupings in transactional sales data.

5.5 Linear Regression (Sales Amount Prediction)

Unlike the resume project where Linear Regression was used on a binary target (inappropriate), here it is correctly applied to predict the continuous Sales dollar amount.

- Features: Quantity, Discount, Year, Month
- Target: Actual Sales ($)

Metric	Value	Interpretation
MAE	~$175–200	Average prediction error per transaction
RMSE	~$300–350	Larger errors exist for high-value outlier transactions
R^2 Score	~0.10–0.16	16% variance explained — sales are non-linear by nature
Conclusion	—	Linear Regression captures broad trends; tree models outperform for classification

6. Results & Conclusions

6.1 KNN Model

- The KNN model with k=5 achieved moderate accuracy (~58–62%) using only Quantity, Discount, Year, and Month to avoid data leakage.
- Feature selection was intentional to test model performance on truly independent variables.
- Confusion matrix showed balanced errors — model avoids heavy bias toward either class.
- Conclusion: KNN is a reasonable baseline for sales classification using temporal and discount features alone.

6.2 Decision Tree Model

- Best-performing model overall: ~90%+ accuracy, F1 ~0.91, ROC AUC ~0.91.
- Category and Region emerged as the strongest decision nodes.
- High recall indicates the model rarely misses actual high-sales transactions.
- The Decision Tree's visual interpretability is a key business advantage — stakeholders can follow the prediction logic directly.
- Conclusion: Decision Tree is the recommended production model for this classification task.

6.3 Logistic Regression

- Single-feature logistic regression (Profit only) produced AUC ~0.5–0.6 — insufficient for reliable classification.
- Multi-feature model improved class separation but class 1 precision (~8%) remains a limitation.
- Logistic Regression serves best as a transparent, interpretable baseline — not the primary prediction engine.

6.4 Key Business Insights

- Technology products consistently drive the highest sales values — prioritizing this category for stocking and promotion is advisable.
- Q4 (October–December) shows peak sales seasonality — inventory and marketing investment should be concentrated in this period.
- West and East regions outperform in total sales; the South region shows opportunity for improvement.
- High-discount rates are associated with lower or negative profit — a discount strategy review is recommended, particularly for Furniture.
- Corporate customers generate above-average transaction values — targeted B2B sales strategies can yield higher revenue per engagement.
- Loss-making transactions are concentrated in specific sub-categories (Tables, Bookcases) — these warrant pricing or cost renegotiation.

7. Dashboard (Using Streamlit)

An interactive **Streamlit dashboard** was developed to provide a business-oriented view of sales performance and profitability. The dashboard enables dynamic filtering and real-time exploration of the Superstore dataset through multiple interactive visualizations. Key dashboard components include:

- **KPI Cards** displaying Total Sales, Total Profit, Total Orders, Units Sold, and Profit Margin for quick performance monitoring.
- **Monthly Sales & Profit Trend Analysis** using interactive line charts to track business performance over time.
- **Category-wise Sales Analysis** through horizontal bar charts highlighting revenue contribution by product category.
- **Region-wise Sales Distribution** using donut charts for comparative regional performance evaluation.
- **Top 10 Products by Sales** visualization to identify the highest revenue-generating products.
- **Sub-Category Sales Breakdown** using treemap visualization for detailed product-level insights.
- **Customer Segment Performance Analysis** comparing sales and profit across Consumer, Corporate, and Home Office segments.
- **Discount vs Profit Analysis** using bubble charts to examine the relationship between discount levels, sales volume, and profitability.
- **Interactive Filters** for Order Date, Region, Category, Sub-Category, Customer Segment, Ship Mode, and State, enabling dynamic data exploration.
- **Raw Data Explorer and CSV Download Feature** allowing users to inspect filtered records and export customized datasets for further analysis.

This dashboard was implemented using **Streamlit, Pandas, and Plotly**, providing an interactive and user-friendly environment for business intelligence and decision-making.

9. References

- [1] Kaggle, 'Datasets — Sample Superstore,' [Online]. Available: https://www.kaggle.com/datasets
- [2] GeeksforGeeks, 'Data Analysis with Python,' [Online]. Available: https://www.geeksforgeeks.org/data-analysis/data-analysis-with-python/
- [3] Scikit-learn Developers, 'sklearn Documentation,' [Online]. Available: https://scikit-learn.org/stable/
- [4] W3Schools, 'Python Machine Learning — KNN,' [Online]. Available: https://www.w3schools.com/python/python_ml_knn.asp
- [5] GeeksforGeeks, 'Decision Tree — Introduction with Example,' [Online]. Available: https://www.geeksforgeeks.org/machine-learning/decision-tree-introduction-example/
- [6] IBM, 'What is Linear Regression?,' [Online]. Available: https://www.ibm.com/think/topics/linear-regression
- [7] GeeksforGeeks, 'Power BI Tutorial,' [Online]. Available: https://www.geeksforgeeks.org/power-bi/power-bi-tutorial/
- [8] GeeksforGeeks, 'A Beginner's Guide to Streamlit,' [Online]. Available: https://www.geeksforgeeks.org/python/a-beginners-guide-to-streamlit/
- [9] Towards Data Science, 'Understanding ROC Curves,' [Online]. Available: https://towardsdatascience.com/understanding-roc-curves
- [10] M. Soliman, 'Data Preprocessing in Machine Learning,' LakeFS Blog, [Online]. Available: https://lakefs.io/blog/data-preprocessing-in-machine-learning/

— End of Report —